A

STM Exercise 9

Phase Diagrams

(Fe-C)

IRON-CARBON (Fe-C) PHASE DIAGRAM

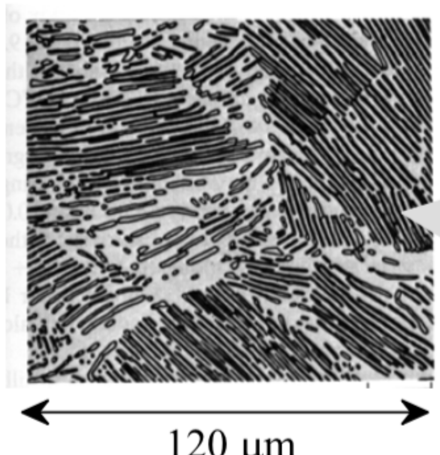
- 2 important points

  -Eutectic (A):

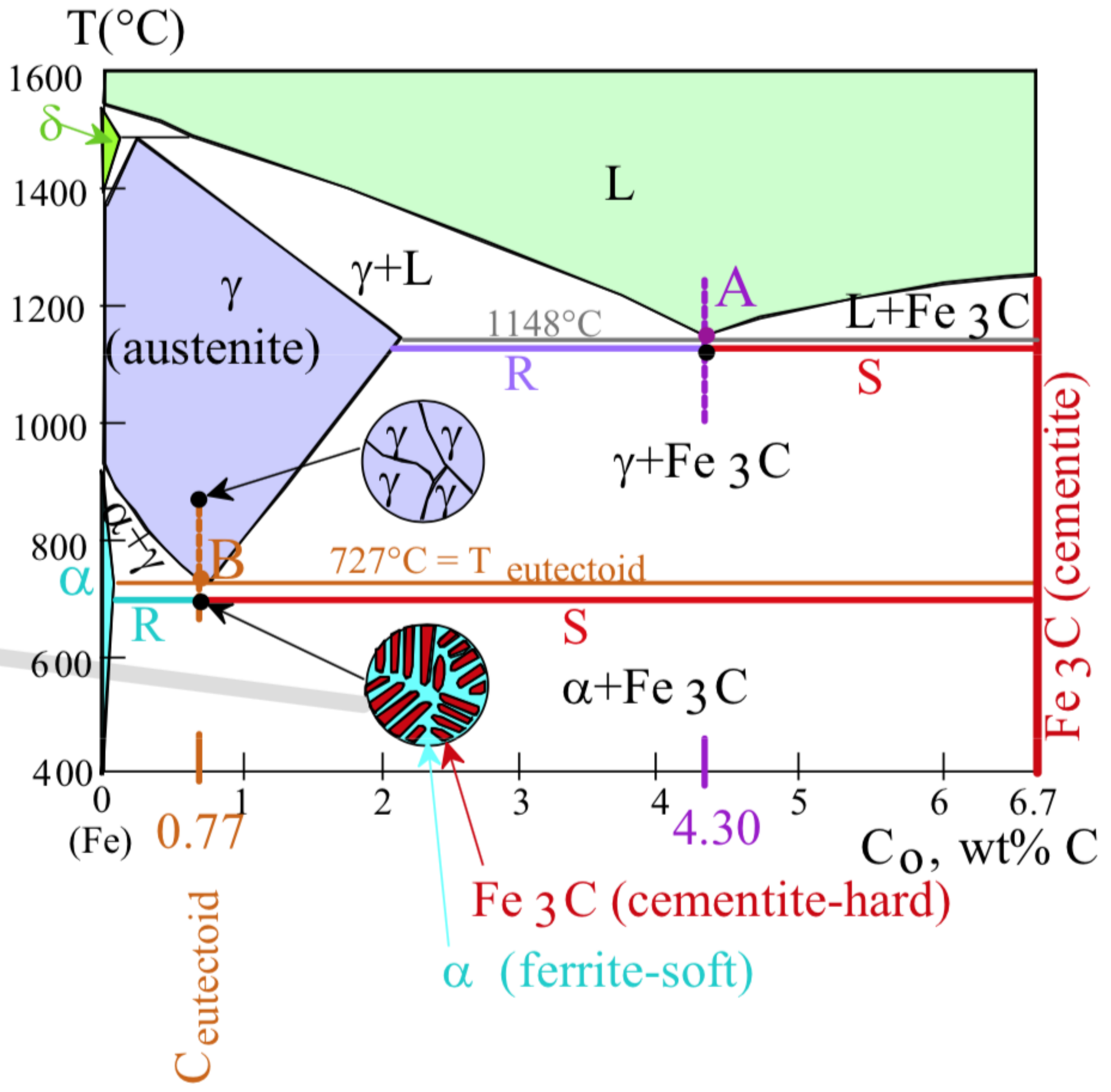
  $L \Rightarrow \gamma + Fe_3C$

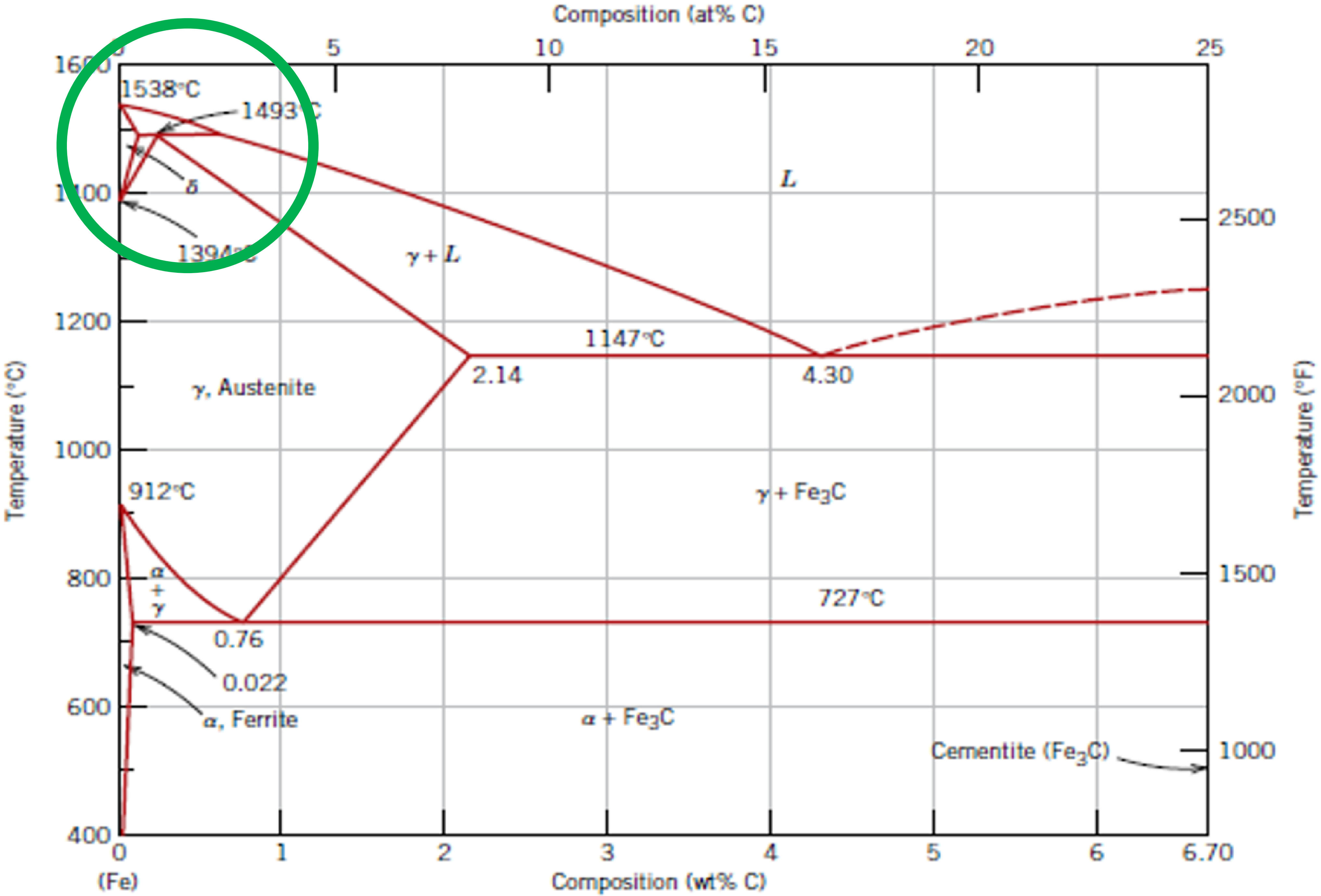
  -Eutectoid (B):

  $\gamma \Rightarrow \alpha + Fe_3C$

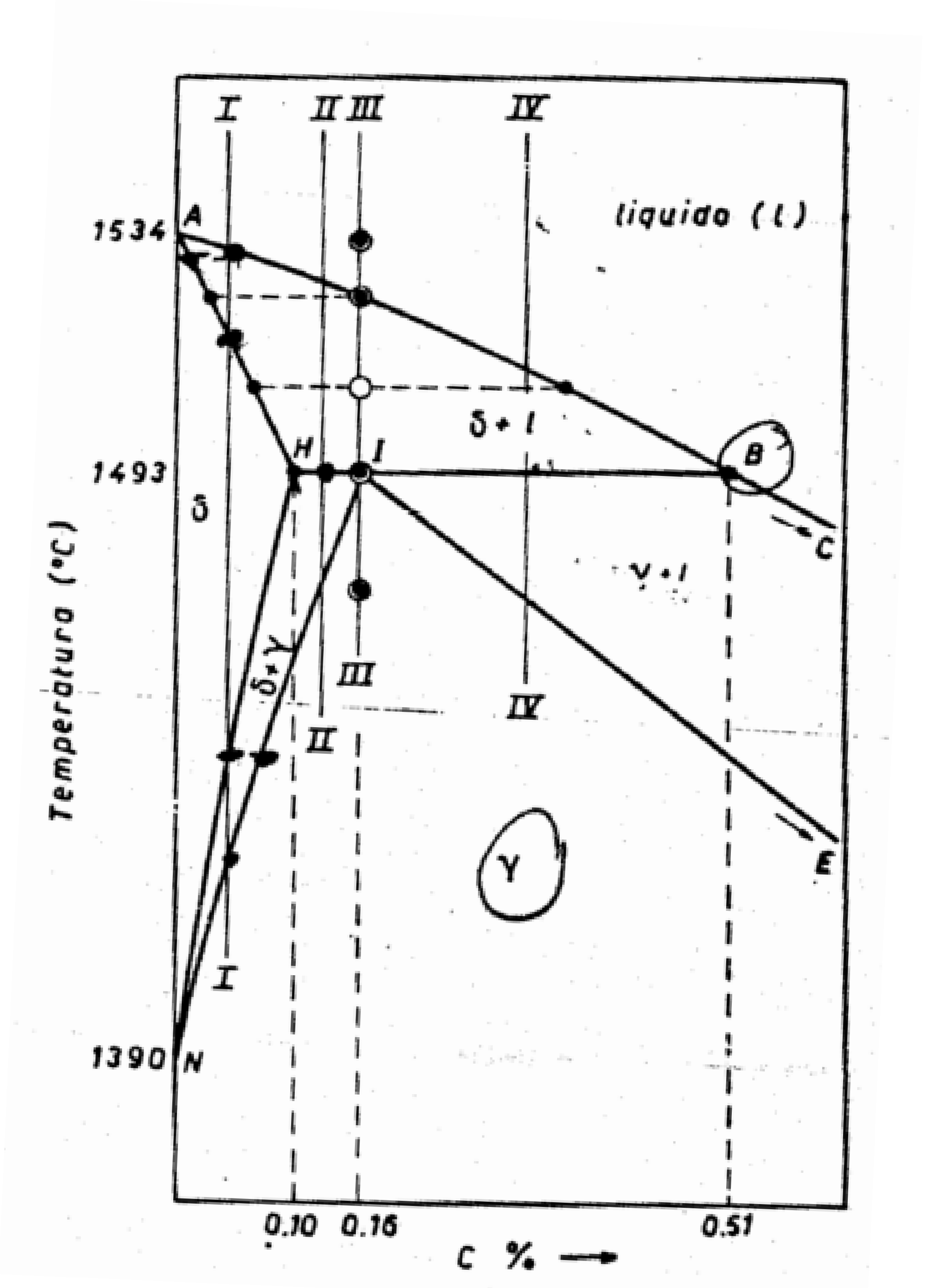


Result: Pearlite = alternating layers of α and Fe_3C phases.

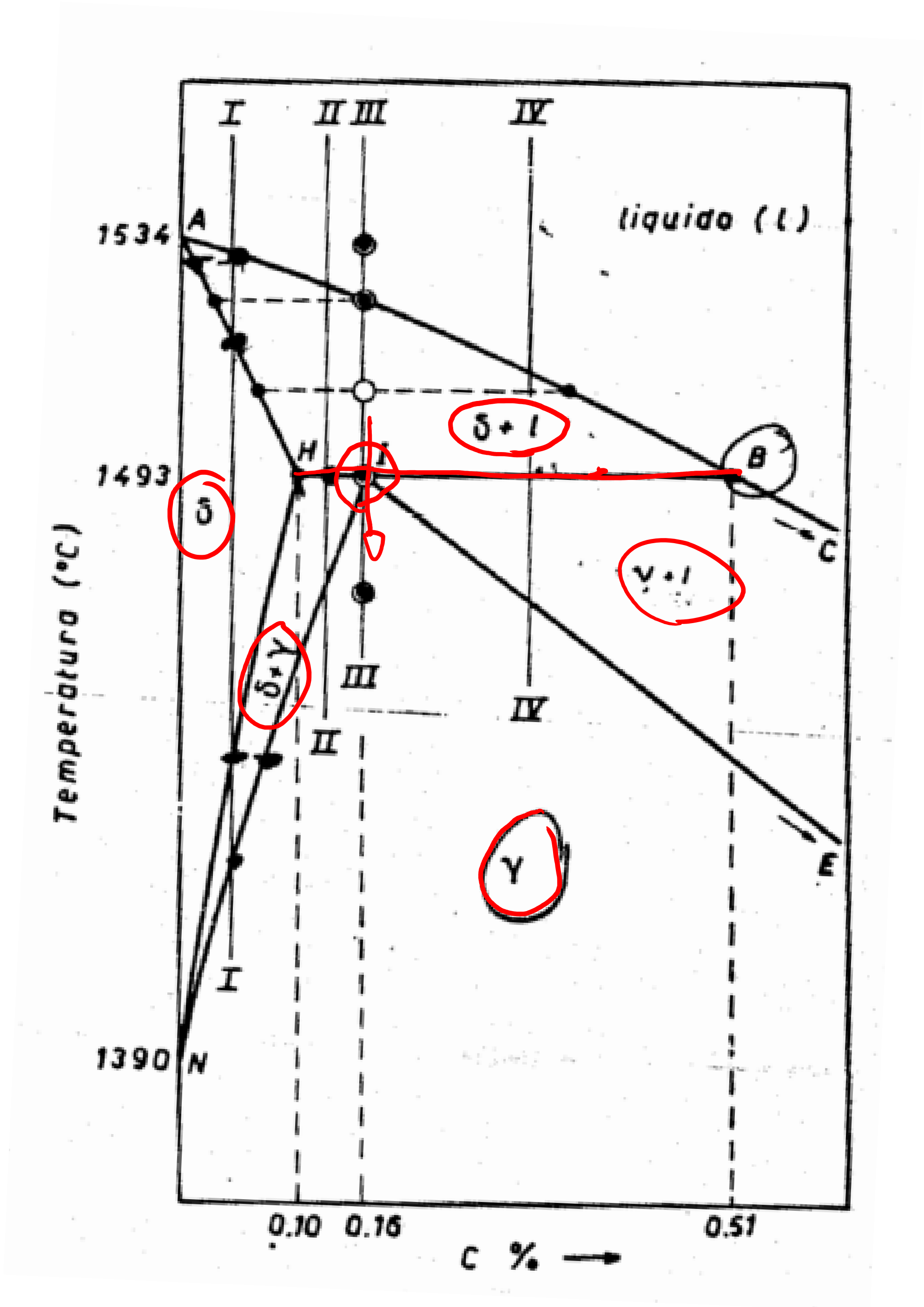




Composition (at% C)
0
5
10
15
20
25
1600
1538°C
1493°C
L
1400
1394°C
δ
γ + L
2500
1200
1147°C
2.14
4.30
γ, Austenite
2000
1000
912°C
γ + Fe3C
800
α
+
γ
727°C
1500
0.76
0.022
600
α, Ferrite
α + Fe3C
Cementite (Fe3C)
1000
400
0
(Fe)
1
2
3
4
5
6
6.70
Composition (wt% C)
Temperature (°C)
Temperature (°F)



Exercise 1: which transformation occurs in point I?



Exercise 1: which transformation occurs in point I?

Point I is a **PERITECTIC** transformation

$\delta + L \leftrightarrow \gamma$

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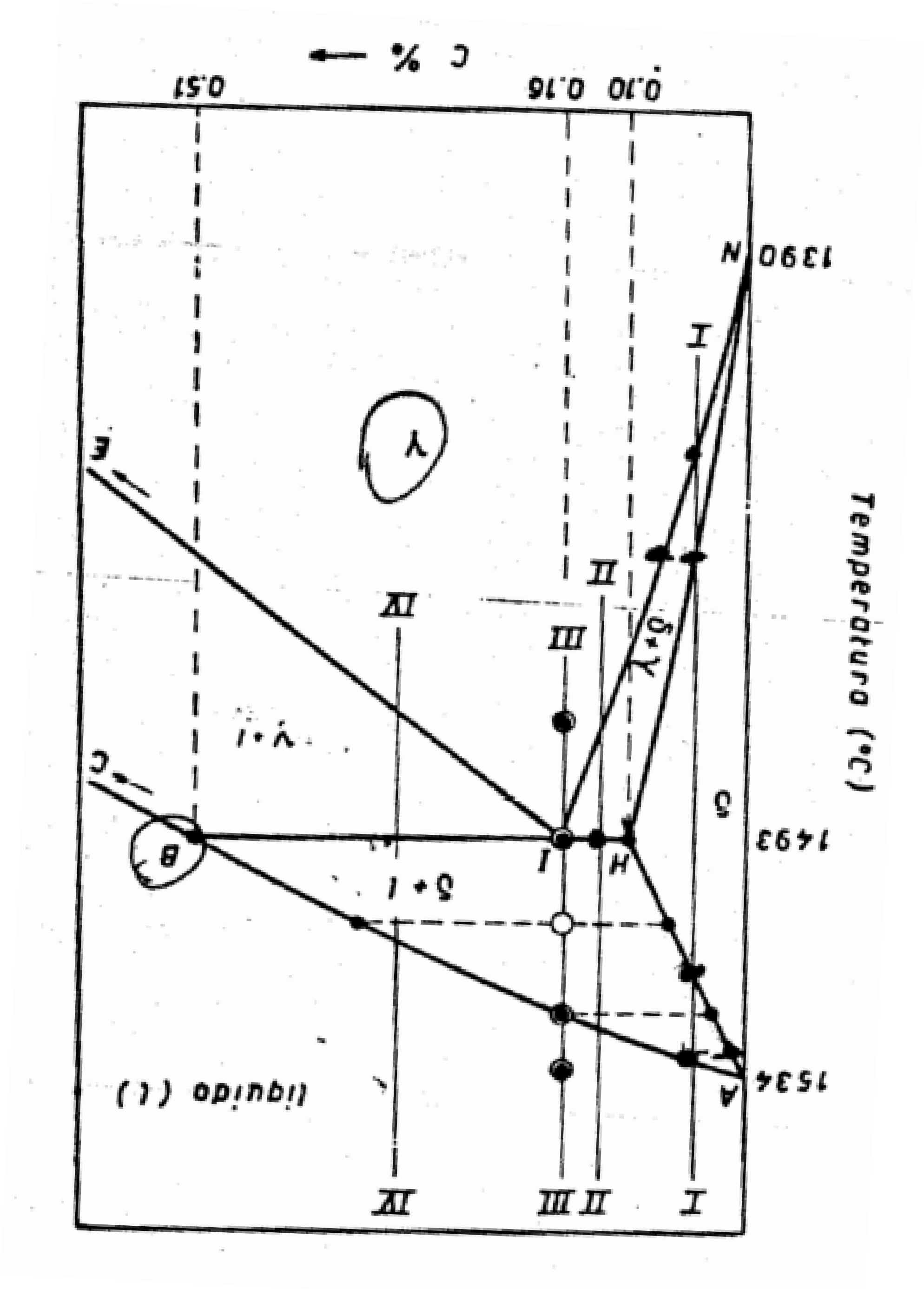
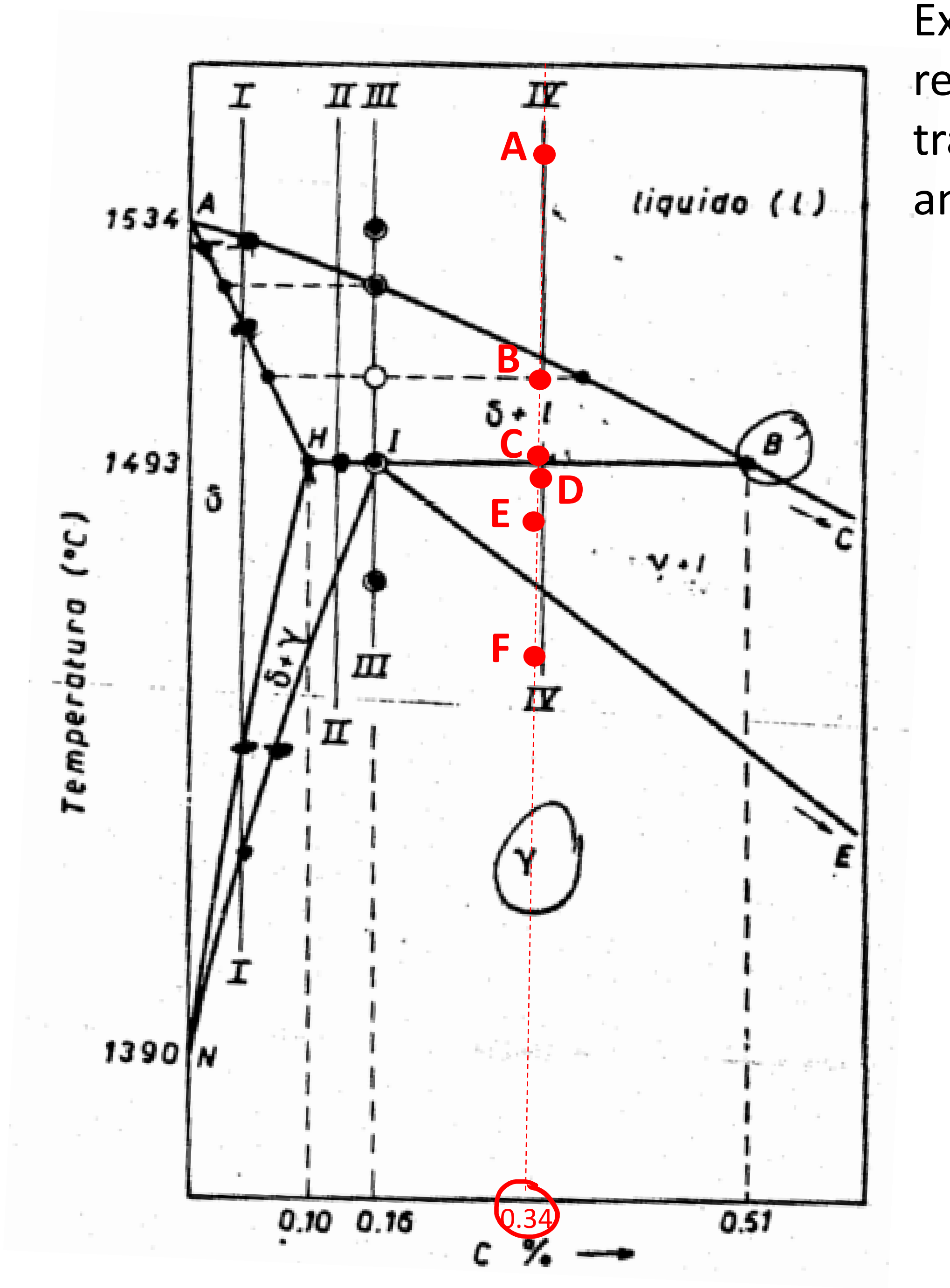


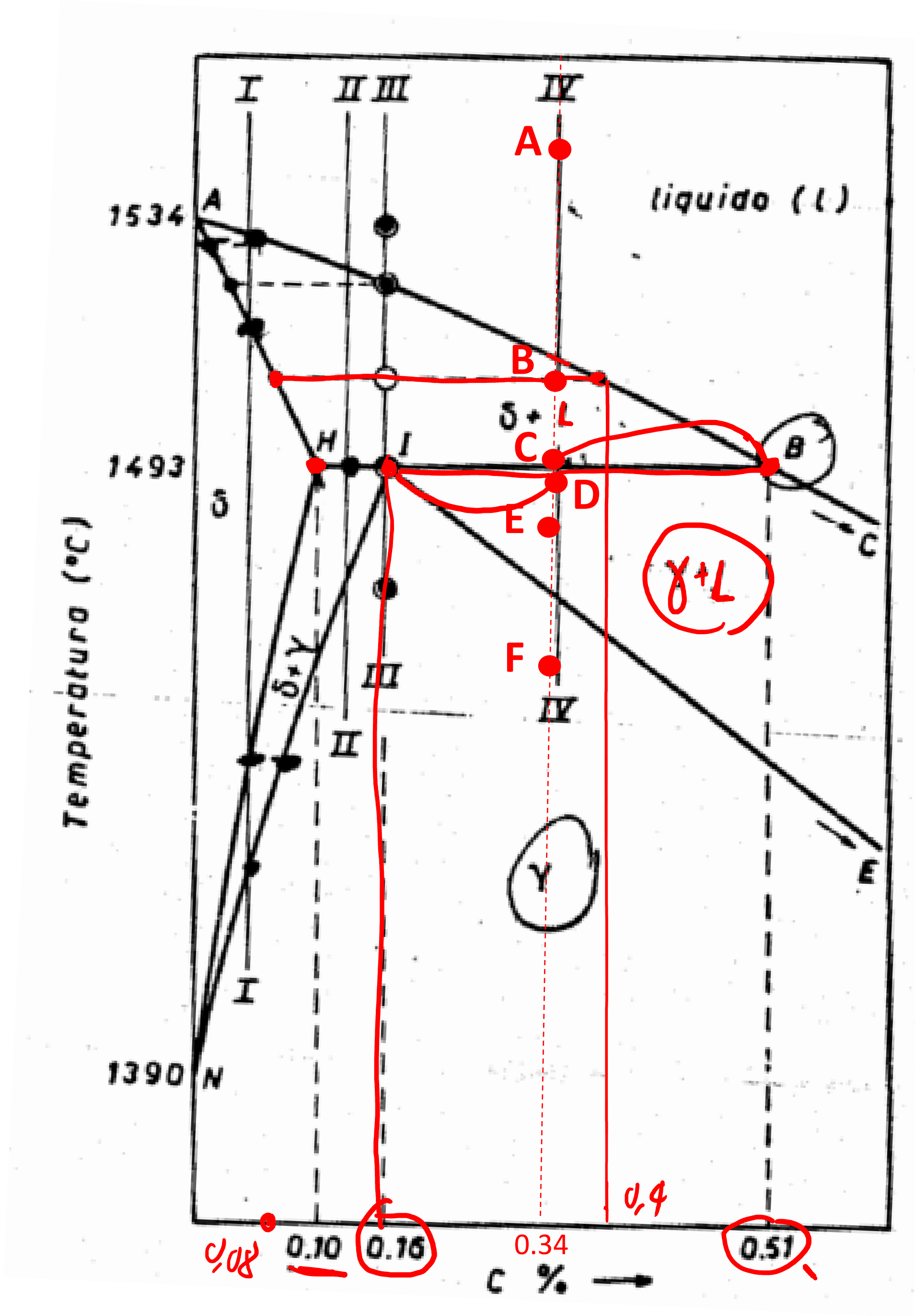
Point I is a **PERITECTIC** transformation

δ+L ↔ γ



Exercise 2: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D (at the end of the peritectic transformation), E and F for a steel with 0.34%C.

$C_0 \to 0{,}34\%C$



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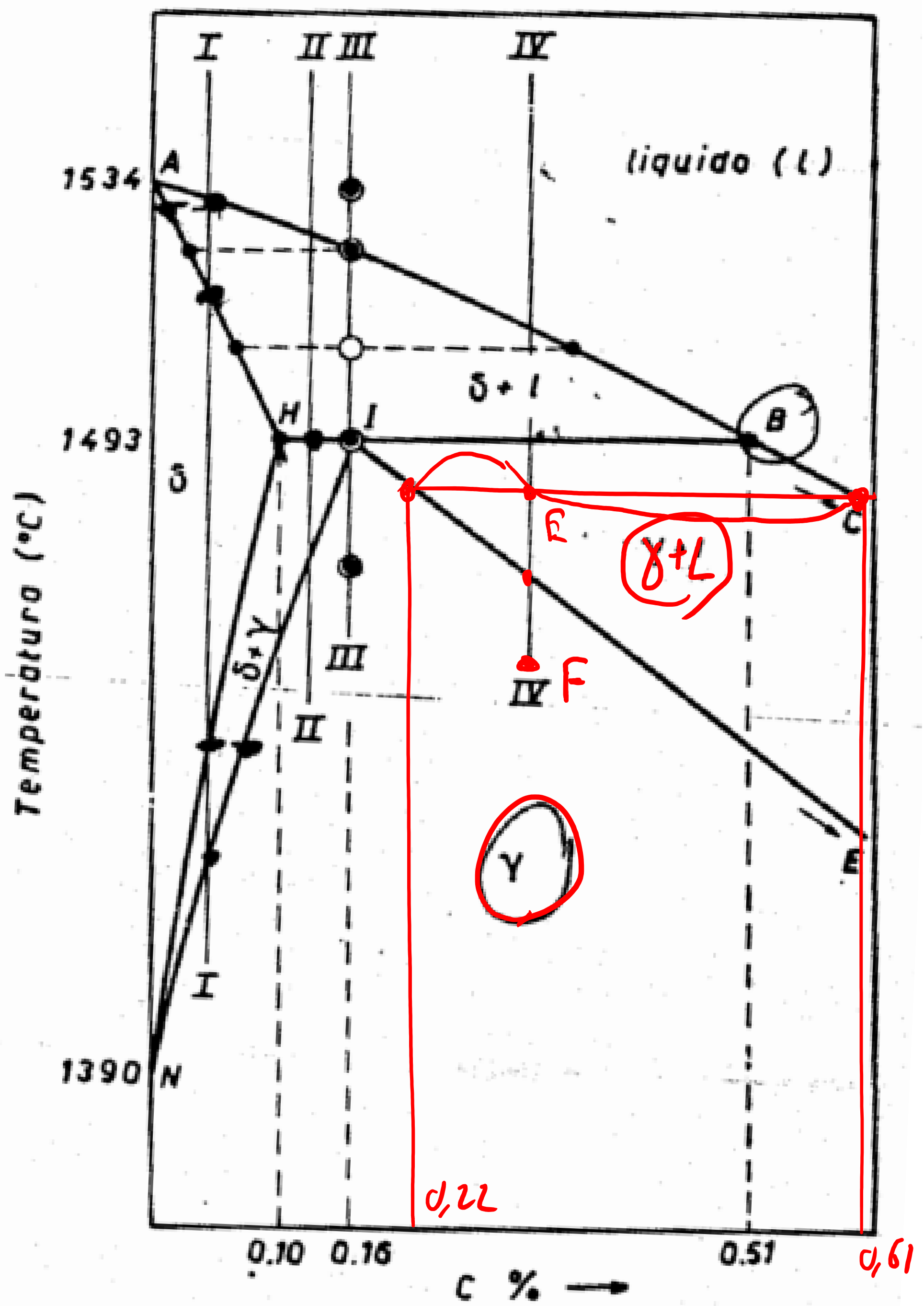
Point (A)	L	
Composition	0,34% C	
Relative amount	100%	

Point (B)	δ	L
Composition	0,08 % C	0,40% C
Relative amount	18,7%	81,3%

Point (C)	δ	L
Composition	0,10 % C	0,51 % C
Relative amount	42%	58%

$$\frac{0,51-0,34}{0,51-0,10} \cdot 100$$

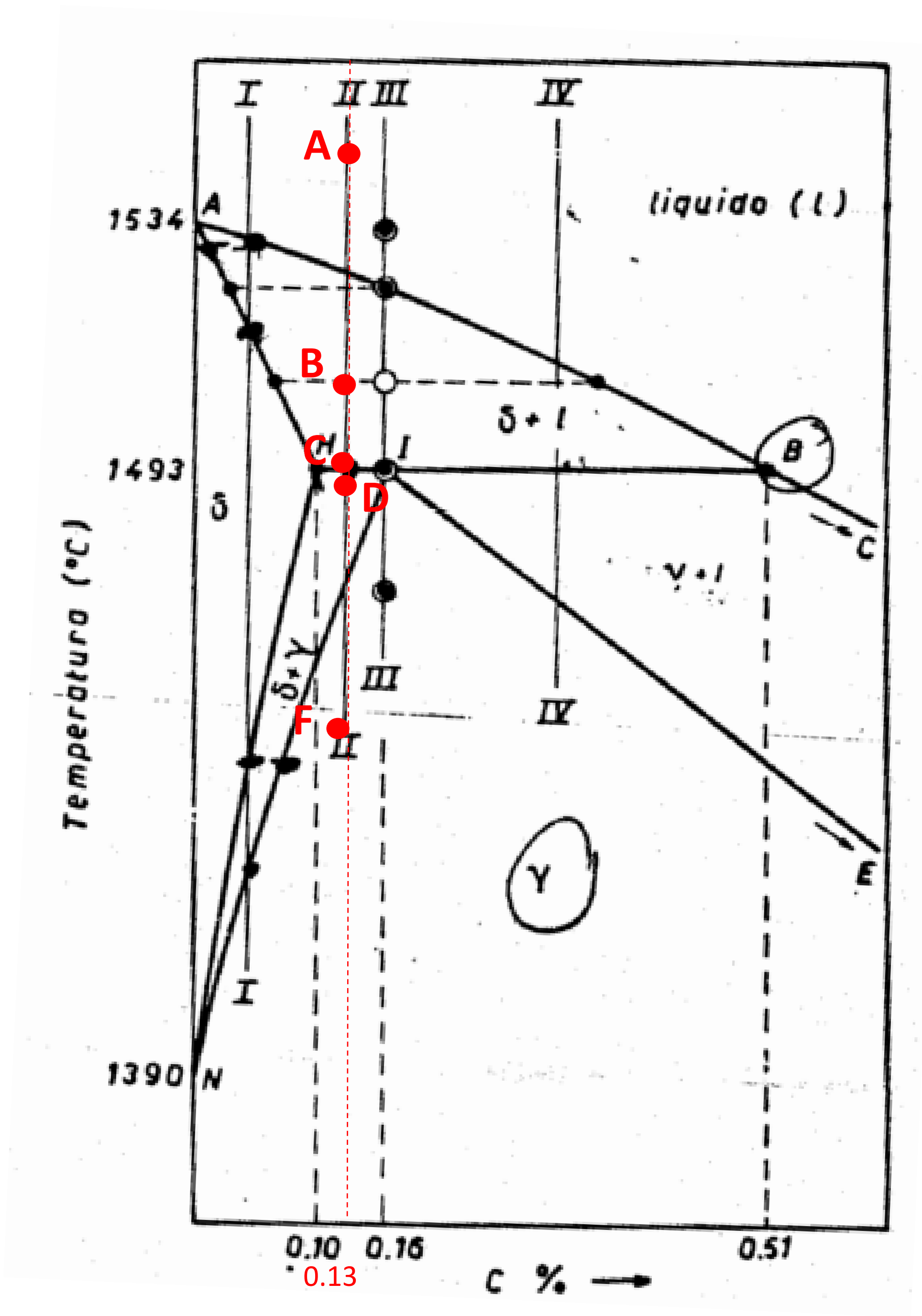
Point (D)	γ	L
Composition	0,16% C	0,51% C
Relative amount	49%	51%



Point (E)	γ	L
Composition	0,22 %C	0,61 %C
Relative amount	69 %	31%

Point (F)	γ	
Composition	0,34 %C	
Relative amount	100 %	

Exercise 3: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D(at the end of the peritectic transformation), ~~E~~ and F for a steel with 0.13%C.



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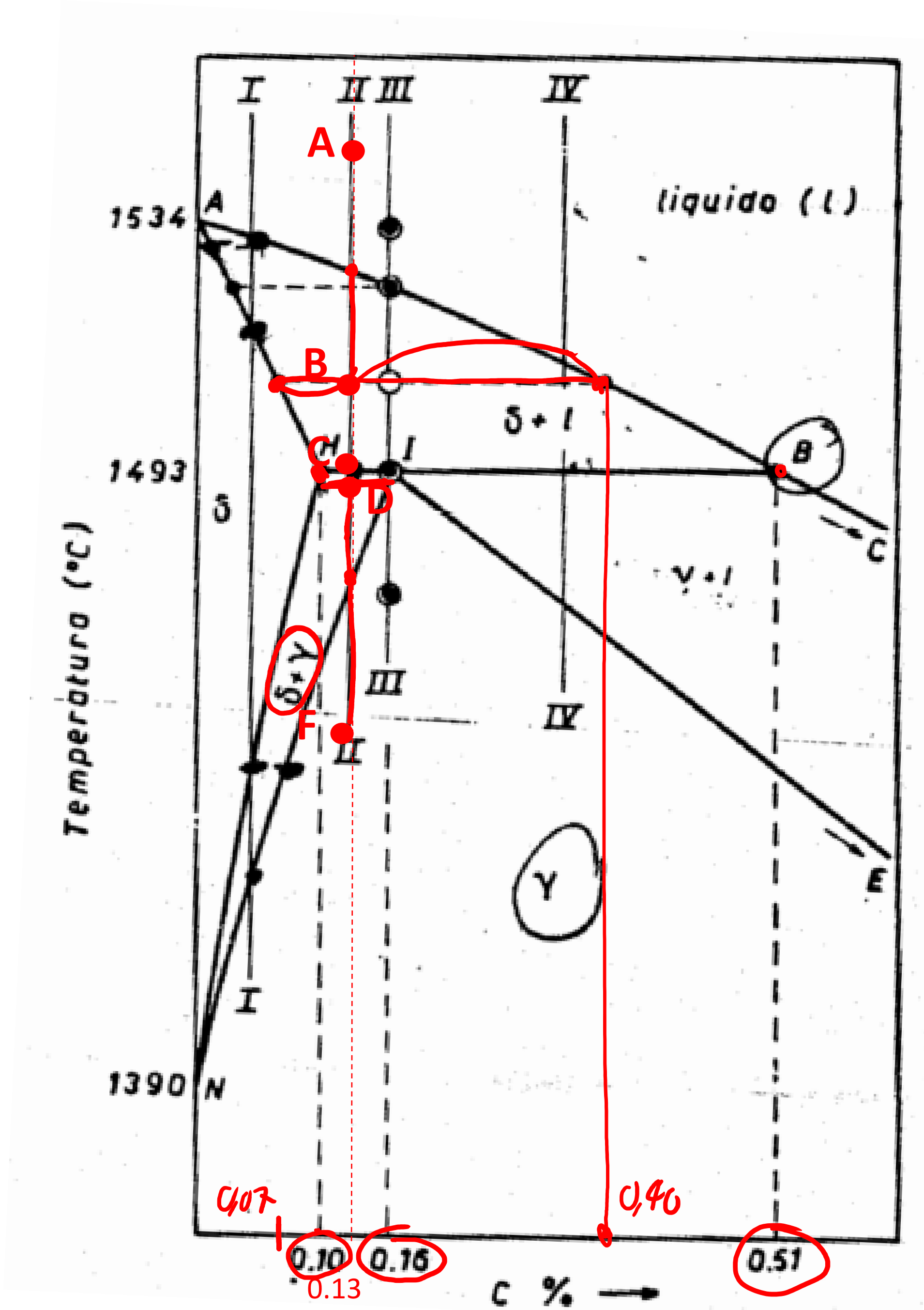


Point (B)	δ	L
Composition	0,07%C	0,40%C
Relative amount	82%	18%

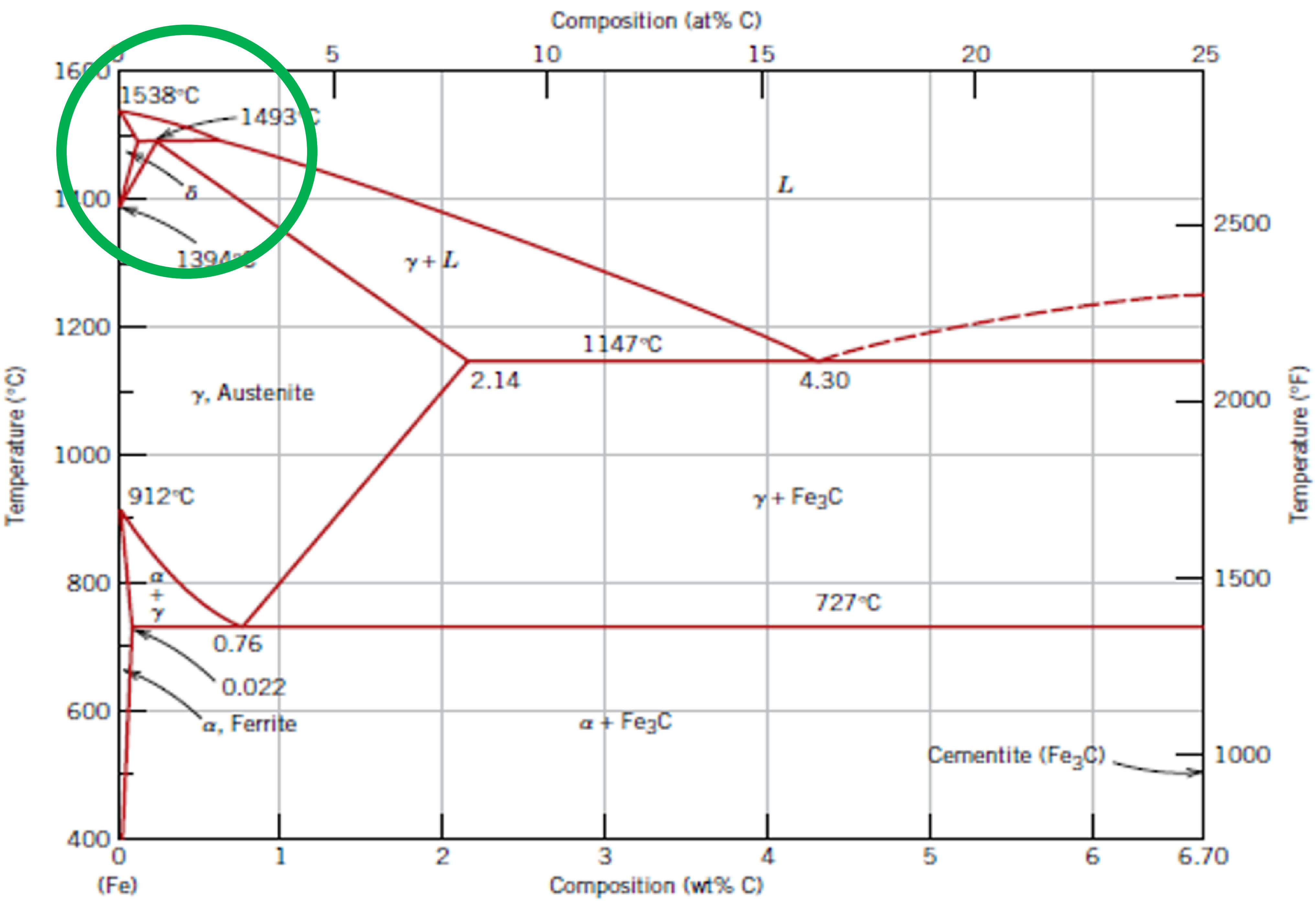
Point (C)	δ	L
Composition	0,10%C	0,51%C
Relative amount	93%	7%

Point (D)	δ	γ
Composition	0,10%C	0,16%C
Relative amount	50%	50%

Point (F)	γ
Composition	0,13%C

END !

Questions ?



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